

00IMO2

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Problem

Let a, b, c be positive real numbers so that $abc = 1$. Prove that

$$\left(a - 1 + \frac{1}{b}\right) \left(b - 1 + \frac{1}{c}\right) \left(c - 1 + \frac{1}{a}\right) \leq 1.$$

Solution. Substitute $a = \frac{p}{q}, b = \frac{q}{r}, c = \frac{r}{p}$. Then it is equivalent to proving,

$$(p + q - r)(q + r - p)(p + r - q) \leq 1$$

Substitute $x = p + q - r, y = q + r - p, z = p + r - q$. Then it is equivalent to proving,

$$8xyz \leq (x + y)(y + z)(z + x)$$

This is of course trivial by AM-GM inequality.